



Data Engineer — Technical Exercise

Take-home brief and dataset

Hg | Internal IT, Platforms & BizOps | Issued September 2026 | Please return within 4 working days

1. About this exercise

Thank you for your time so far. This exercise is the second stage of our process. It is a short, practical piece of work on a small synthetic dataset, shaped like the data you would work with in the role. We use it to see how you think about data quality, how you design a model, how you write for other people, and how you use AI tools in your engineering work.

Time. We have designed this for **1–2 hours**. Please treat that as a real limit rather than a target to beat. Depth of judgement matters more to us than polish, and at the panel we will ask you what you would have done with more time — a thoughtful answer to that question is worth more than an extra hour of formatting. Please tell us roughly how long you spent.

What we are looking for. The panel scores the same seven things for every candidate: data-quality instinct, modelling judgement, SQL / dbt fluency, quality of specification writing, critical thinking and the questions you ask back, how you explain your choices, and how deliberately you use AI. There is no trick — the tasks below map straight onto that list.

What happens next. Your submission goes to the panel a day before we meet. At the panel (60 minutes) you will walk us through your work for about 15 minutes, then we will discuss it and work through some related scenarios you will not have seen before, with ~10 minutes at the end for your questions.

2. The scenario

Hg's finance team collects a small set of monthly KPIs from each portfolio company through a standard template. Companies submit roughly a week after month end into a portfolio-monitoring tool, and a monthly extract from that tool is landed in **Mantle**, Hg's internal enterprise data platform (AWS, Terraform, dbt, Snowflake).

The investment team has asked for a **Portfolio KPI dashboard in Sigma** built on top of Mantle (business requirements in section 5). Today the equivalent view is a manually assembled spreadsheet. You have been asked to take the raw extract and design the path from source to dashboard — and, because part of the ingestion may be built by an external managed-service partner, to write the specification they would build from.

Everything in the dataset is fictional. Companies, funds, figures and FX rates are all synthetic. Nothing here describes a real Hg portfolio company, so you are free to load the files into any tool, including AI tools, without any confidentiality concern.

3. What you have been given

File	What it is	Rows
companies.csv	Master list of portfolio companies: identifiers, fund, sector, reporting currency, investment status.	9
kpi_monthly.csv	Monthly KPI submissions — revenue, ARR, gross churn, headcount — as received by the monitoring tool, September 2024 to August 2026.	203

File	What it is	Rows
fx_rates_monthly.csv	Month-end FX rates to GBP for the currencies in use.	71
data_dictionary.md	Column-level definitions for the three files (also reproduced in Appendix A).	—

The extract is exactly as it came out of the tool. Like any real extract, it is not perfect — noticing where and deciding what to do about it is part of the exercise.

4. The tasks

Six short tasks. The time guides add up to a little under two hours; they are there to help you allocate effort, not to be tracked. Any format is fine for your answers — a document, a notebook or a small repo.

Task 1 — Profile the data (≈30 min)

List the data-quality issues you find and, for each, how you would detect it **automatically** in a pipeline (a dbt test, a query, a rule — whatever you would actually use). Then tell us which issues you would **block** the load on and which you would **flag** and let through, and why.

Please be specific: which rows, which companies, which months. A prioritised list beats an exhaustive one.

Task 2 — Design the model (≈30 min)

Sketch a dbt layering — staging → intermediate → mart — that would serve the Portfolio KPI dashboard described in section 5, in Sigma. Name the key models and state the **grain** of each. Note the tests you would add at each layer and what failure each test protects against.

A diagram, a bullet list or a short table is fine. We are interested in your reasoning about grain, where each kind of cleansing belongs, and how the dashboard's requirements shape the mart.

Task 3 — Write the code (≈20 min)

Two or three SQL / dbt snippets: at least one **cleansing step** from Task 1 and one **mart-level model** from Task 2. Working code is preferred (Snowflake dialect ideally, but any SQL engine you can run is fine — the files are small). Pseudo-code is acceptable if the logic is clear. Please tell us what you actually ran.

Task 4 — Write the spec (≈20 min)

About half a page: the **ingestion specification** you would hand to an external managed-service partner to build the source-to-staging load for this extract. Include acceptance criteria that can be tested, and describe how you would QA their delivery before signing it off.

Assume the partner is competent but has never seen this data or spoken to the finance team. Write for them.

Task 5 — Questions back (≈5 min)

Five questions you would ask the business before building any of this for real.

Task 6 — AI leverage note (≈10 min)

Hg is an AI-forward firm and we expect you to use whatever AI tools you normally would on this exercise. Please tell us:

- Which tools you used, and for what.
- Where the output was wrong or needed checking, and what you did about it.
- Which parts you deliberately did yourself, and why.

This is assessed on judgement, not volume. Not using AI is not an advantage; using it uncritically is a disadvantage. We will ask you about at least one piece of AI-assisted work at the panel.

5. Portfolio KPI dashboard — business requirements

High-level brief from the investment team and Portfolio Operations. Deliberately not a full specification: use it to shape your model in Task 2, and challenge it in Task 5.

Purpose and users

A single monthly view of operating performance across the portfolio, replacing a spreadsheet that Portfolio Operations currently assembles by hand around the tenth working day after month end. Around 40 users: deal partners and investment teams (read), Portfolio Operations and Finance (read and investigate).

Questions it must answer

- How is the portfolio — and each fund — tracking this month on ARR, revenue, churn and headcount, versus last month and versus twelve months ago?
- Which companies moved the most, up or down, and on which metric?
- Which companies have not yet reported for the selected month?
- What has changed since we last looked — has any prior month been restated, and by how much?

Metrics

Metric	Definition (as the business describes it)
ARR	Annual recurring revenue at month end. Shown in GBP; local currency available on drill-down. Growth month-on-month and year-on-year.
Revenue	Monthly recognised revenue and last-twelve-months revenue, GBP.
Gross churn	Gross monthly revenue churn, as a percentage. Portfolio and fund figures should be ARR-weighted, not simple averages.
Headcount	FTE at month end; revenue per head as a derived measure.
Reporting status	For each company and month: reported, not yet reported, missing, or reported but not yet convertible (e.g. no FX rate).

Rules and constraints

- **Reporting currency is GBP**, converted at the month-end rate for the month being reported. Local-currency values must remain visible. A constant-currency growth view is a nice-to-have, not a requirement for the first release.
- **Hierarchy and filters**: portfolio → fund → company, with a month selector and 24 months of history. Exited companies should appear in history up to their exit, not after.
- **No silent gaps**. Where data is missing, late or has been adjusted by the pipeline (for example an inferred currency or a corrected unit), the dashboard must show that rather than a blank or a guessed number.
- **Restatements are expected**. Companies re-submit prior months from time to time. Users want the latest view by default, with the fact of a restatement — and its size — visible.
- **Refresh**: monthly, once the extract lands (target: dashboard current by the twelfth working day), with the ability to re-run when a late submission arrives.
- **Access**: all investment staff may see all companies for the first release; no row-level security is required yet.

Out of scope for the first release

Budgets and forecasts, LP reporting, company-level P&L detail, and anything below monthly grain.

6. Submitting your work

- **Format:** anything that is easy to read — a document (PDF or Word), a notebook, or a link to a small repository. Please keep SQL in text form (not screenshots) so we can run it.
- **Contents:** your answers to Tasks 1–6, the approximate time you spent, and anything you would want the panel to know before we meet.
- **Deadline:** within four working days of receiving this brief, via La Fosse, who will forward it to the panel. If anything in the brief is unclear, send your question through La Fosse and we will come back to you quickly.
- **At the panel:** prepare a ~15-minute walkthrough of your submission. Slides are optional; talking through your document is fine.

We appreciate the time this takes. We keep the exercise deliberately small so that the conversation at the panel can be about your judgement rather than your stamina — and we look forward to it.

Appendix A — Data dictionary

Three CSV files, UTF-8, comma-separated, header row, ISO dates. Extract taken 10 September 2026. All content is synthetic.

companies.csv — one row per portfolio company

Column	Type	Description
company_id	text	Stable internal identifier, PC - nnn.
company_code	text	Three-letter short code used by the finance team in KPI templates.
company_name	text	Legal / trading name.
fund	text	Fund that holds the investment.
sector	text	Sub-sector label.
hq_country	text	Country of headquarters.
reporting_currency	text	Currency the company reports its management accounts in (ISO 4217).
investment_date	date	Date of initial investment.
status	text	Active or Exited.
exit_date	date	Populated only when status = Exited.

kpi_monthly.csv — monthly KPIs as submitted by each company

Grain as supplied: one row per company per month per extract. Companies submit roughly 6–10 days after month end via a finance template; the monitoring tool appends each submission it receives.

Column	Type	Description
company_code	text	Short code — joins to companies . company_code.
period_month	text	Reporting month, YYYY-MM.
period_end_date	date	Last calendar day of the reporting month.
currency	text	Currency of the monetary columns on this row (ISO 4217).
revenue	number	Total recognised revenue for the month, in currency, whole units (not thousands).
arr	number	Annual recurring revenue at month end, in currency, whole units.
gross_churn_pct	number	Gross monthly revenue churn, in percentage points (e.g. 1.2 = 1.2 %).
headcount	integer	Full-time-equivalent employees at month end.
source_extract_date	date	Date the row was received by the monitoring tool.

fx_rates_monthly.csv — month-end FX rates to GBP

Column	Type	Description
month	text	YYYY-MM.
from_ccy	text	Source currency (ISO 4217).
to_ccy	text	Always GBP.
rate	number	GBP per 1 unit of from_ccy (e.g. 0.79 means USD 1 = GBP 0.79).
rate_source	text	Where the rate came from.